

A Path-Backtracking-based Trust Management Scheme for VANETs

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Abstract—Trust management is a representative research domain to ensure secure communication and cooperation among entities in Vehicular Ad-hoc Networks (VANETs). However, vehicles as entities are not always trustworthy because of the openness characteristics of VANETs, such as various malicious participants, and selfish behaviors. In this paper, we propose a Path-Backtracking-based Trust Management Scheme (PBTMS) for VANETs to evaluate the trustworthiness of entities based on the interaction histories (according to previous interaction counts), message path-backtracking (tracking the origin and route of messages), and multi-path analysis (analyzing the likelihood of malicious vehicle presence). Specifically, within the PBTMS framework, the trustworthiness of vehicles is meticulously evaluated. By incorporating both entity trust and path trust metrics, PBTMS computes an aggregate trust value for each vehicle. Extensive simulations demonstrate that PBTMS surpasses other baseline algorithms in accuracy, achieving higher precision in evaluating entity trustworthiness.

Index Terms—Trust management, VANETs, Path-Backtracking

I. INTRODUCTION

Vehicular Ad-hoc Networks (VANETs) significantly influence the development of the Intelligent Transport System (ITS). As a revolutionary approach to transportation, ITS autonomously organizes networks for high-safety travel, and environmental protection. It leverages technologies such as wireless communication and sensors to improve communication among vehicles, infrastructure, drivers, and pedestrians.

Given their high mobility and dynamic connections, VANETs are vulnerable to a variety of threats. These threats include external attacks originating from outside the network. Additionally, internal attacks originating from malicious entities within the network, such as data interception and the spread of false information. Previous security solutions in VANETs have emphasized cryptographic methods [1] to ensure secure and confidential communication. Cryptographic methods primarily serve the purpose of ensuring data confidentiality and integrity. However, they are unable to combat the threat of internal attacks. For instance, an attacker with valid credentials and verified identity may tamper with the message content, potentially leading to traffic congestion or accidents [2].

Addressing the issue of internal attacks, trust management models have emerged as a significantly effective solution. The

trustworthiness of vehicles is determined by examining their interaction histories. The primary focus is placed on the quality of information exchange and consistent behaviors, rather than merely the frequency of interactions.

Existing trust management solutions in VANETs have shown notable limitations. Firstly, they cannot adapt well to the dynamical topology of VANETs because they focus on frequent interactions and raw data exchanges, but fail to trace the transmission path of messages. Secondly, several solutions only defend against single type of attacks. Nevertheless, sophisticated attacks break through their defense and make networks vulnerable. Thirdly, both the reliance on static parameters and the lack of effective mechanisms to trace malicious behaviors of vehicles are weaknesses of previous solutions. To address the above problems, we introduce the Path-Backtracking-based Trust Management Scheme (PBTMS), considering the crucial need for enhanced transmission path tracking. This scheme effectively identifies sources of malicious activities. The main contributions of our work are summarized as follows:

(1) Previous works focus on interaction histories but overlook the subtle details of message transmission paths, causing potential inaccuracies in identifying and isolating malicious vehicles. PBTMS is conceived to afford a more detailed consideration of both historical interaction records and message transmission paths. By analyzing interaction histories and incorporating path-backtracking, PBTMS accurately evaluates vehicle trustworthiness, ensuring improved security and dependability for all network entities.

(2) Additionally, in order to better distinguish between malicious and trustworthy vehicles, PBTMS utilizes a method based on statistical analysis for trust threshold updating, thereby increasing the reliability and security of VANETs.

II. RELATED WORKS

A. Entity-Centric Trust Models

Entity-centric trust models are designed to evaluate the trustworthiness of entities (vehicles) and ensure secure communication in VANETs. The work [3] proposed a strategy that formulates an aggregated score for vehicles, combining their trust level, cooperative behavior, and last hop recommendation. The vehicle with the highest score is then designated for data

forwarding. Furthermore, Xia *et al.* [4] advanced a robust trust inference mechanism for VANETs, notably impervious to black/grey-hole incursions. The architecture of this scheme combines subjective trust, based on historical vehicular interactions, with recommendation trust sourced from adjacent vehicular nodes' consensus. Although entity-centric trust models identify malicious vehicles effectively, they struggle to detect multiple vehicle collaboration attacks.

B. Data-Centric Trust Models

Data-centric trust models evaluate the quality of data exchanged between vehicles. The work [5] proposed a trust model that utilizes data quality as a primary factor in determining trust values. The model highlights that trust values should focus on data quality, rather than the direct vehicle trust. The work [6] introduced a Context-aware Trust Management Framework (CTMF). This framework utilizes context information to enhance the accuracy of trust evaluation. However, data-centric trust models are effective in detecting attacks that involve malicious data, but fail to detect attacks that involve malicious vehicles. This limitation arises primarily because these models are intrinsically designed to assess the integrity of the data itself, rather than the behavioral patterns or intentions of the vehicles transmitting data.

C. Hybrid Trust Models

Hybrid trust models utilize data and vehicle behaviors for a comprehensive trust score calculation in VANETs. Placzek and Bernas [7], innovatively amalgamate behavior-centric and similarity parameters to forge a sophisticated hybrid trust management framework for VANETs. This intricate design primarily targets the mitigation of malevolent data intrusions, notably from Sybil attacks. Trust scores are derived by comparing projected and actual vehicular behaviors, particularly in response to traffic signals. Additionally, they consider the consistency in messages from neighboring vehicles. The work [8] introduced a novel hybrid trust management scheme. Differing from traditional models, this scheme evaluates trustworthiness in two steps: first by assessing the node at the transport layer, and then by computing the trustworthiness of the data at the application layer.

D. Motivation

Based on the above considerations, we note that several trust management schemes in VANETs neglect the dynamic nature of vehicular transmission paths. Most of them focus on static thresholds that cannot adapt to various scenarios, thus resulting in inefficiencies and inaccuracies of malicious vehicle identification. In contrast, PBTMS addresses above limitations, providing a more adaptive trust evaluation tailored to the dynamic conditions of VANETs.

III. SYSTEM AND ATTACK MODEL

A. System Model

Fig. 1 shows the system model of PBTMS.

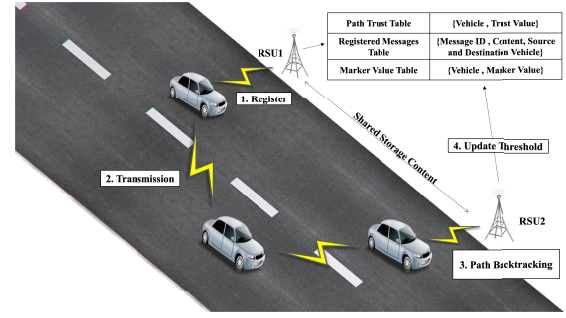


Fig. 1. The System Model of PBTMS.

- **Vehicles:** A vehicle establishes connections with Road Side Units (RSUs) and other vehicles to transmit messages. Vehicles serve as both the source and destination of messages. Furthermore, with the assistance of RSUs, a vehicle evaluates the trustworthiness of other vehicles.
- **RSUs:** RSUs are responsible for calculating the path trust, in addition to verifying the integrity of message transmissions while also confirming their authenticity. If any malicious behavior is detected, the path will be traced back to identify the malicious vehicles. Furthermore, RSUs periodically exchange stored contents in their memories, such as the path trust table, registered message table, and marker value table.

B. Attack Model

Three attack models are utilized to evaluate the performance of PBTMS, including Man-In-The-Middle (MITM), black hole, and on-off attacks.

- **MITM attacks:** By hijacking and modifying messages, the attacker can manipulate the information flow.
- **Black hole attacks:** Compromised vehicle within the network, known as the black hole, drops all messages received from other vehicles.
- **On-off attacks:** Malicious vehicles with the on-off attack model alternate their behaviors. They switch between benign and malicious actions to evade detection.

IV. A PATH-BACKTRACKING-BASED TRUST MANAGEMENT SCHEME

A. Overview

In PBTMS, it is assumed that RSUs are completely trustworthy, while the trustworthiness of vehicles remains uncertain. The overview of PBTMS is illustrated in Fig. 2.

(1) Vehicles communicate with RSUs after generating messages, providing evidence for message authentication during the message registration process. Only registered messages can be transmitted to others. Each message consists of identity (ID), content, source, and destination. Once each message is received, the receiver records the source and content of the message. This information is utilized to identify malicious vehicles during the path-backtracking process.

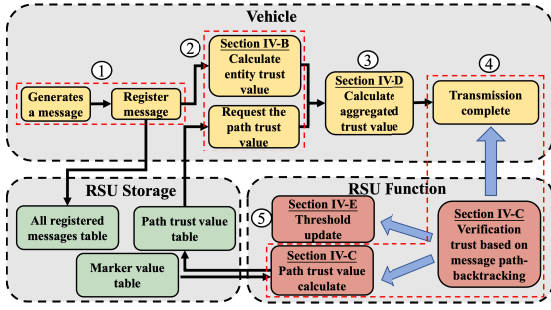


Fig. 2. System overview of PBTMS.

TABLE I
ABBREVIATIONS LIST.

Terms	Description
V_i	The i -th vehicle
Msg_i	The i -th message
$T_E^{V_{ij}}$	The entity trust value of V_j obtained from V_i
t_{ijk}	The temporal decay factor from the view of V_i to V_j
F_i	Marker Value of V_i
$T_P^{V_i}$	Path trust value of V_i
$P_B^{V_i}$	Verification trust of V_i
$P_M^{V_i}$	Multi-path trust of V_i
$P_F^{V_i}$	Marker trust of V_i
T_{V_i}	Aggregate trust value of V_i
T_{thr}	Trust threshold

(2) The message-sending vehicle calculates the entity trust value of the message receiver vehicle. Then, it requests the path trust value of the message receiver from the RSUs.

(3) The aggregate trust of the message-receiving vehicle is calculated based on the entity and path trust value. This process continues until the message is delivered to its destination.

(4) If a transmission path does not exhibit any malicious behavior, all vehicles on that path will receive path trust values, these values are recorded in the path trust value table of the RSUs. However, if the RSUs detect a message verification failure, path-backtracking is initiated to identify malicious vehicles, and their trust values will be decreased. To prevent MITM or on-off attacks, PBTMS tags malicious vehicles with a marker value during the process of path-backtracking and logs their misconduct.

(5) To enhance malicious vehicle detection, PBTMS utilizes a dynamic trust threshold update. This mechanism distinguishes malicious from legitimate vehicles by adjusting trust thresholds based on trust value statistics.

B. Entity Trust Value Calculation

Table I lists important abbreviations. According to the interaction histories of vehicles, the entity trust value serves as a crucial metric to assess the trustworthiness of vehicles. Specifically, to calculate the entity trust value from V_i to V_j , the following formula is applied:

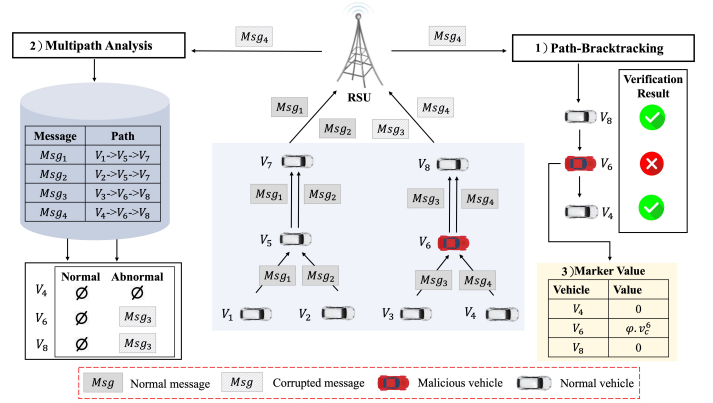


Fig. 3. The process of path trust evaluation.

$$T_E^{V_{ij}} = \frac{\sum_{k=1}^n w_k \cdot s_{ijk} \cdot t_{ijk}}{\sum_{k=1}^n w_k \cdot \sum_{k=1}^n s_{ijk} \cdot t_{ijk}}, \quad (1)$$

where w_k represents the weight of V_k , which indicates that V_k has interacted with V_j within a specified time window. The weight w_k is assigned for the aggregate trust value (discussed in Section IV-D) of V_k , while n denotes the total number of vehicles. s_{ijk} denotes the number of times that V_i and V_j have forwarded messages through V_k . t_{ijk} represents the temporal decay factor, to ensure that more recent interactions have a greater impact on the trust value, while past interactions have a smaller impact. This is achieved by multiplying the interaction count, with the temporal decay factor. The temporal decay factor can be defined as follows:

$$t_{ijk} = e^{-\lambda(t_{now} - t_k)}, \quad (2)$$

where t_{now} represents the current time, t_k denotes the latest interaction time between V_k and V_j . λ represents the time decay coefficient. Here, $\lambda = 0.01$.

C. Path Trust Value Calculation

Path trust ensures the integrity of message transmission among vehicles. Upon message delivery to its destination, the content is cross-checked against registered data to identify discrepancies. A mismatch indicates potential tampering, prompting trust evaluation and subsequent storage of the path trust value in the RSUs' vehicle path trust value table. If altered message contents are detected by RSUs, a path-backtracking mechanism is activated. Path-backtracking mechanism scrutinizes both message content and communication routes for inconsistencies. Here, each vehicle receiving the message records both the identity of the sender and the content of the message, allowing for the identification of the tampering source. The process of path trust evaluation is shown as Fig. 3.

1) *Verification trust based on message path-backtracking:* Each vehicle records messages received from others and provides evidence for path-backtracking mechanism. This mechanism is designed to trace the origin and path of each message, ensuring its integrity and authenticity. As shown in Fig. 3,

Msg_4 received by the 8-th vehicle (denoted as V_8) differs from the message registered by V_4 at the RSU, indicating that it has been tampered with. This tampering is attributed to V_6 . However, the message received by V_6 is consistent with that registered by V_4 at the RSU. Therefore, V_4 and V_8 are verified, and malicious behaviors of V_6 are detected. As a result of the tampering by V_6 , its malicious detection count increases ($v_f^6 + 1$). Meanwhile, the benign detection counts of V_4 and V_8 increases (i.e., $v_c^4 + 1$ and $v_c^8 + 1$ respectively). In this context, v_f^i and v_c^i are utilized to evaluate the trust value of vehicles: v_f^i counts how many times a vehicle has been verified as malicious, while v_c^i denotes the number of times a vehicle has successfully passed verification checks. The verification trust of V_i is then expressed as:

$$P_B^{V_i} = \begin{cases} \frac{v_c^i}{N_{Ver}^i}, & \text{if } N_{Ver}^i > 0, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

where N_{Ver}^i represents the total number of verifications. If $N_{Ver}^i = 0$, then the verification trust value is 0. This is because some malicious vehicles might conduct black hole attacks.

2) *Multi-path trust based on vehicle multi-path analysis:* Malicious vehicles might attack continuously or intermittently. For instance, as shown in Fig. 3, Msg_4 is believed to be tampered with. This conclusion is drawn after comparing the delivered message with its registered content, revealing discrepancies. According to the detection of enrolled messages, V_4 does not transmit any other tampered message, while V_6 and V_8 have transmitted the tampered messages Msg_3 . Then, V_6 and V_8 have each transmitted tampered messages twice, whereas V_4 has only done so once in contrast. Therefore, V_6 and V_8 are considered as high risk vehicles. The multi-path trust of V_i is expressed as:

$$P_M^{V_i} = \begin{cases} \frac{N_s^{V_i} - N_f^{V_i}}{N_s^{V_i} + N_f^{V_i}}, & \text{if } N_s > N_f, \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

where $N_s^{V_i}$ represents the number of successfully relayed normal messages in V_i , and $N_f^{V_i}$ denotes the number of successfully relayed tampered messages in V_i .

3) *Marker trust derived from marker values:* When the RSU detects a malicious vehicle through the path-backtracking mechanism, it increases the marker value of malicious vehicles to keep track of their malicious behaviors. This marker value mechanism serves as a cumulative record of a vehicle's behavior over time. The incremental value of the marker value for the malicious vehicle is expressed as follows:

$$F_i^{new} = F_i^{old} + \varphi \cdot v_f^i, \quad (5)$$

where F_i^{old} represents the marker value of V_i , and φ is a penalty factor, $0 < \varphi < 1$. Here, $\varphi = 0.5$. The penalty factor, in this case, determines the severity of the penalty applied to the vehicle's marker value. A higher value of φ would mean a steeper penalty for malicious actions. When a vehicle has been involved in a few malicious activities, its marker value will

increase with each detected malicious action. As the marker value approaches 1, the vehicle is closer to being identified as malicious. Once the marker value reaches or exceeds 1, the vehicle is deemed malicious.

Given this clear identification of malicious intent, it becomes imperative to enforce strict punitive measures to uphold both the security and trustworthiness of vehicles within the VANETs. When a vehicle is identified as malicious, its aggregate trust value (discussed in Section IV-D) is set to 0. It ensures the malicious vehicle is effectively removed from networks and cannot operate malicious behaviors, such as generating false messages. Finally, the marker trust of V_i is expressed as:

$$P_F^{V_i} = \frac{1}{e^{F_i^{new}}}. \quad (6)$$

Eq. (6) further emphasizes the inverse relationship between the marker value and the trustworthiness of a vehicle. As the marker value increases, the trustworthiness of the vehicles with more malicious behaviors decreases exponentially.

Ultimately, the path trust value is obtained by weighting the verification trust, multi-path trust, and marker trust. α and β are weight parameters with $0 < \alpha, \beta < 1$, and they are set by the evaluation system. Here, $\alpha = 0.4$ and $\beta = 0.3$. The computation for the path trust value of V_i is expressed as:

$$T_P^{V_i} = \alpha P_B^{V_i} + \beta P_M^{V_i} + (1 - \alpha - \beta) P_F^{V_i}. \quad (7)$$

Algorithm 1: Path Trust Value Calculation

Input: Message Msg , Message path Msg_{path} ,
Registered message Msg_{reg} ,
Message-receiving vehicle V_j

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1 if  $V_j$  is destination vehicle then
2   if  $Msg$  differs from  $Msg_{reg}$  then
3     Trigger path-backtracking;
4     for each vehicle  $V$  in  $Msg_{path}$  do
5       Update  $N_f^V$ ;
6       if  $Msg$  from  $V$  differs from  $Msg_{reg}$  then
7         Update  $F_i^{new}$  for  $V$  and adjust
          aggregate trust value if  $F_i^{new} > 1$ ;
8   else
9     Update  $N_s^V$ ;
10  for each vehicle  $V$  in  $Msg_{path}$  do
11    Compute  $T_P^V$  using Eqs.(3) to (7);

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In PBTMS, we assume that vehicles do not alter message transmission paths. Since ensuring the authenticity of the paths may require the assistance of cryptography, it is not within the scope of this paper. The overall process for path trust calculation is illustrated in Algorithm 1.

D. Aggregate Trust Value Calculation

Based on the entity trust value and path trust value, the aggregated trust value can be calculated by a weighted approach. $\delta \in (0, 1)$ is a weight parameter. For instance, $\delta = 0.5$ indicates that entity trust and path trust are equally important to calculate. According to Eq. (1) and Eq. (7) the aggregate trust value of V_i is expressed as:

$$T_{V_i} = \delta T_E^{V_{ij}} + (1 - \delta) T_P^{V_j}. \quad (8)$$

E. Trust Threshold Update

The proposed trust threshold update algorithm calculates the trust threshold based on the InterQuartile Range (IQR) and median of aggregate trust values. The trust threshold update algorithm employs a series of steps to determine the trust threshold for vehicles. It begins with the collection of trust values by the RSUs and sorts them in ascending order.

Firstly, the median and quartiles (the first and third quartiles denoted as Q1 and Q3) of the trust values are calculated to determine the IQR. This statistical approach is crucial as it provides a more comprehensive understanding of the trust values' distribution, ensuring that outliers or extreme values do not skew the threshold. Secondly, based on the derived IQR and the median of all aggregated trust values from vehicles, the system recalibrates the threshold value. The updated formula of the threshold value is shown as follows:

$$T_{thr} = median - (IQR \times \gamma), \quad (9)$$

where $IQR = Q3 - Q1$. γ is a multiple parameter. Here, $\gamma = 1$. However, when transmission interaction among vehicles is insufficient, vehicles have low trust threshold. To address this issue, the minimum value of threshold is set to 0.1. This ensures that even in scenarios with limited interactions, the trust evaluation remains robust. Based on the above considerations, the proposed trust threshold update algorithm offers a reliable and adaptive approach to computing the trust threshold.

V. EXPERIMENTS

A. Simulation Setup and Evaluation Metric

The performance of PBTMS is evaluated through simulation experiments based on the Opportunity Network Environment simulator [9]. The experimental scenario is modeled after the physical layout of Helsinki city, covering an area of $4500m \times 3400m$ square meters, as shown in Fig.4. Furthermore, 100 vehicles and 12 RSUs are randomly instantiated within the confines of the digital map. The vehicles navigate towards pre-determined endpoints by following the shortest path computed by the Dijkstra algorithm [10]. Moreover, the transmission range, transmission rate, and buffer size are set at $100m$, $900kbps$, and $10MB$, respectively. The simulation time was set to 2000s, with a warm-up time of 200s before the start time of simulation [11].

In the experiment, we compared PBTMS with IWOT-V [12]. IWOT-V utilizes a Bayesian approach to construct an implicit web, distinguishing between malicious and benign

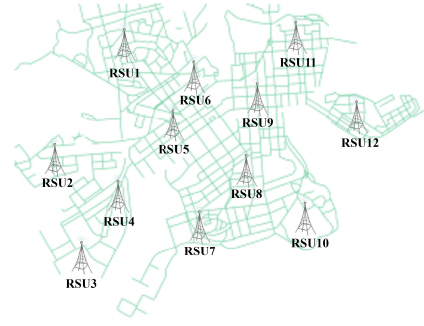


Fig. 4. Simulation scenario of Helsinki City.

vehicles by amalgamating local trust assessments into global trust values. Notably, the two models presented in this study rely on the spray and wait routing protocol¹ [13].

B. Evaluation Metric

The performance of PBTMS is assessed through three metrics: precision, recall, and F-measure. Precision represents the ratio of true positives to all identified positives, while recall captures the proportion of true positives to all actual positives. F-measure, a balance between precision and recall, is the harmonic mean of the two, computed as $\frac{2 * (Precision * Recall)}{Precision + Recall}$.

C. Results and Analysis

Fig. 5(a), the precision metric's trend for PBTMS is particularly noteworthy. With up to 50% malicious vehicles, PBTMS consistently maintains a precision of 1, indicating its impeccable ability to correctly classify malicious vehicles without any false positives. It represents the robustness of the PBTMS.

Fig. 5(b) delves into the recall metric. PBTMS showcases an exemplary performance up to 50% malicious vehicles. However, a discernible dip is observed when the proportion of malicious vehicles touches 40%. This could be due to the sheer volume of malicious activities overwhelming the detection mechanisms.

Fig. 5(c) presents the F-measure for PBTMS, showcasing its outstanding performance once more. PBTMS consistently outperforms the comparative scheme, especially in scenarios with a higher percentage of malicious vehicles. Comparatively, it's evident that PBTMS outperforms IWOT-V in terms of robustness, especially in scenarios with a lower malicious vehicles presence in the VANETs.

In Fig. 6, the aggregated trust values of 100 vehicles in PBTMS with varying weights δ are presented. Vehicles 74-99 are identified as malicious, with specific subsets engaging in on-off, MITM, and black hole attacks. Here, dash lines denote trust thresholds for different weights. Notably, all 25 malicious vehicles were detected under each weight scenario,

¹The Spray and Wait routing protocol is an efficient way to send messages in intermittently connected networks. It works by sending multiple copies of a message and opportunistically forwarding them through the network until the message is received by the destination node.

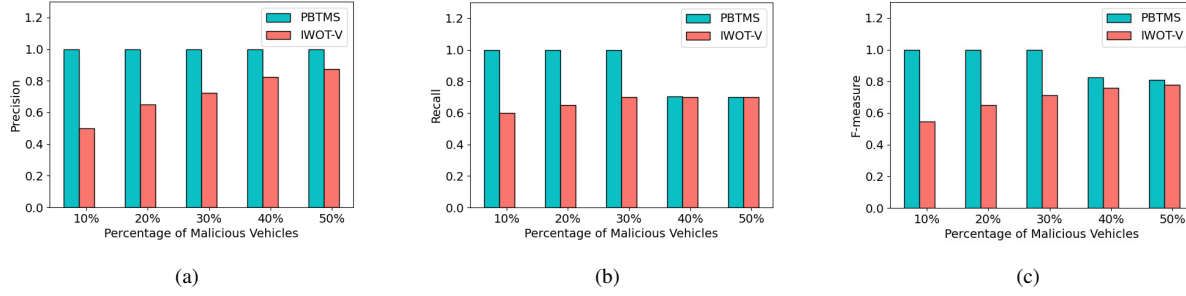


Fig. 5. Three metrics of IWOT-V and PBTMS under different percentages of malicious vehicles. (a) Precision vs percentage of malicious vehicles. (b) Recall vs percentage of malicious vehicles. (c) F-measure vs percentage of malicious vehicles.

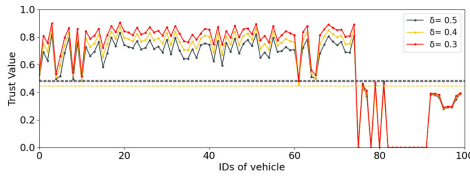


Fig. 6. The aggregated trust values of 100 vehicles with different weights δ in PBTMS.

with their trust values consistently below the respective thresholds. Several malicious vehicles, having launched multiple attacks, saw their trust values drop to zero due to their marker values reaching or surpassing 1. This underscores PBTMS's efficacy in pinpointing multi-attack vehicles and ensuring their persistently low trust scores. Nevertheless, as path trust weight increases, most vehicle trust values also elevate, except for those (91-99) utilizing black hole attacks—these vehicles cannot accrue path trust due to failed message transmissions, leading to diminished aggregate trust when entity trust weight decreases.

VI. CONCLUSION

In this paper, a trust management scheme for VANETs called PBTMS is presented with the aim of improving the reliability of VANETs. In PBTMS, the entity trust value for each vehicle is derived from its interaction histories, with a decay factor incorporated to prioritize recent interactions. The scheme incorporates a path-backtracking mechanism to identify malicious vehicles. The multi-path analysis further boosts the trust value of well-behaved ones, effectively safeguarding against MITM attacks. To counteract on-off attacks and deter vehicles from repeatedly launching assaults, marker values are employed. Additionally, a statistical-based trust threshold updating method is also designed to effectively distinguish between malicious and benign vehicles. Simulation experiments demonstrate the effectiveness and stability of the trust management scheme.

VII. ACKNOWLEDGEMENT

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